

Quiz 5 - Turing Machines

Taken on: October 26, 2024

Total: 5 + 5 bonus points

Question 1

[5 Points]

Give an implementation-level description of a Turing machine that decides the following language:

$\{w \mid w \text{ contains twice as many 0s as 1s}\}$

over the alphabet $\{0, 1\}$.

Question 2: Bonus**[5 Points]**

Give an implementation-level description of a Turing machine that decides the following language:

$$\{w \mid w \text{ does not contain twice as many 0s as 1s}\}$$

over the alphabet $\{0, 1\}$.